

Design and Implementation of a Buck Converter for DC Voltage Regulation

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April 28, 2026

Introduction

A buck converter is a type of DC–DC converter used to step down a higher DC voltage to a lower level efficiently.

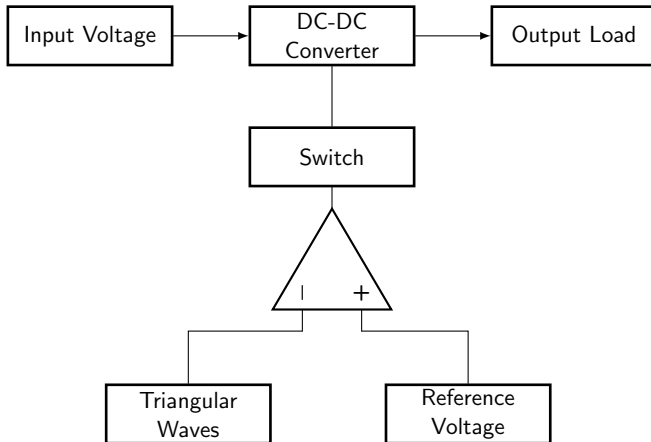
- Widely used for voltage regulation in electronic devices
- Uses switching techniques for high efficiency
- PWM generated using op-amp comparator and triangle wave
- Output voltage controlled by duty cycle

Device Specifications

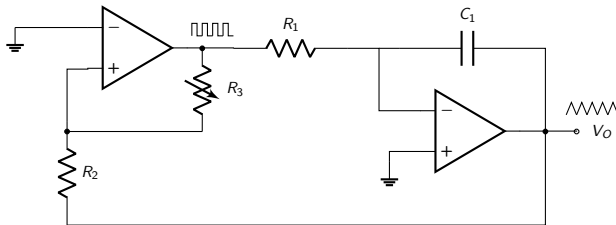
Specifications

- Input Voltage: 15 V
- Output Voltage: 3V – 12 V
- Switching Frequency: 16 kHz
- Maximum Load Current: 1 A
- Duty Cycle Range: $D = 0.2$ to 0.8
- Target Output Ripple Voltage : 1 % of V_{out}
- Target Inductor Ripple Current : 10 % of I_{out}
- Maximum output power : 15 W

Block Diagram



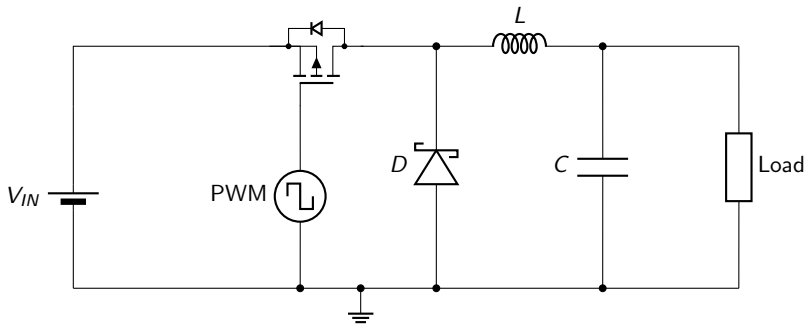
Triangular Waveform Generator



Operation

- Generated using an op-amp integrator and comparator
- Produces a continuous triangular waveform
- Used as carrier signal for PWM generation
- Compared with reference voltage to create PWM output

Buck Converter



Buck Converter Circuit

Circuit Operation

- Input DC voltage V_{in} is applied to MOSFET switch
- PWM signal controls ON/OFF operation of the switch
- Inductor stores energy during ON state
- Diode provides current path during OFF state
- LC filter smooths output and reduces ripple
- Output voltage depends on PWM duty cycle

Op-Amp Selection

1. Slew Rate Requirement To generate square wave in PWM generator circuit,

Minimum Required Slew Rate

$$SR \geq |V_p \cdot j\omega e^{j\omega t}| = 7.5 \times 2\pi \times 16 \times 10^3 \times 10 = 7.5 \text{ V } \mu\text{s}^{-1}$$

2. Gain Bandwidth Requirement To accurately reproduce a square wave, the op-amp must support higher frequency harmonics.

Minimum Required GBW

$$GBW \geq n \cdot f_0 = 10 \times 15 \text{ kHz} = 150 \text{ kHz}$$

Note: Higher-order harmonics are required to approximate a square wave, which increases the effective slew rate requirement. In this design, up to the 10th harmonic is considered.

Op-Amp Selection (Continued)

3. Input Voltage Requirement $V_{CC}^+ - V_{CC}^- \leq 15\text{ V}$

Selected Op-Amp : NE5532

- High slew rate (Typical $9\text{ V } \mu\text{s}^{-1}$).
- 6 to 40 supply voltage range.
- High gain bandwidth product (10 MHz).
- Low cost and readily available.



Switching Transistor Selection

1. **Voltage Rating** Higher than the input voltage, typically 1.2 to 1.5 (22.5 V).
2. **Current Rating** Should be able to handle peak current, so at least 1 A (2 A is recommended for safety).
3. **Switching Voltage:** 0 to -15 V

Selected MOSFET: IRF9540

- **Voltage Rating:** $V_{DS} = 100\text{ V}$ ($> 22.5\text{ V}$)
- **Current Rating:** $I_D = 19\text{ A}$ ($>> 2\text{ A}$)
- **Gate-source threshold voltage** -2 to -4 V.

Diode Selection

- 1. Voltage Rating:** Higher than the input voltage, typically 1.2 to $1.5 \times V_{in}$ (22.5 V).
- 2. Current Rating:** Should handle load current, at least 1 A.
- 3. Fast Switching:** Required to reduce switching losses in PWM operation.

Selected Diode: 1N5819 (Schottky)

- **Reverse Voltage Rating:** $V_{RRM} = 40\text{ V}$ ($> 22.5\text{ V}$)
- **Forward Current Rating:** $I_F = 1\text{ A}$
- **Forward Voltage Drop:** $V_F \approx 0.45\text{ V}$ to 0.6 V (Lower than standard 0.7 V for higher efficiency)

Inductor and Capacitor Selection

Inductor Selection

1. **Inductance:** Based on ripple current requirement
2. **Current Rating:** $I_L \geq I_{peak}$

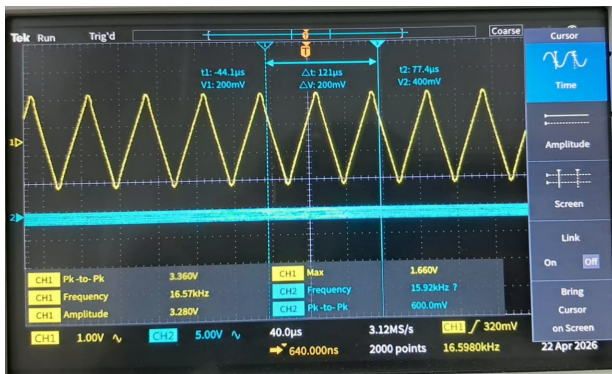
Capacitor Selection

1. **Capacitance:** Based on output voltage ripple
2. **Low ESR:** Reduces ripple

Selected Inductor and Capacitor

- $L = 2.2 \text{ mH}$
- $C = 33 \mu\text{F}$

Triangular Waveform Generation



- The obtained frequency was 16.57 kHz. (< 20 kHz)
- Actual $V_{pp} = 3.36$ V. (< 4 V)

PWM Signal Generation



- $V_{pp} = 13.6 \text{ V}$ and $f = 16.54 \text{ kHz}$.
- Practical Op-Amp slew rate $= \frac{13.52 \text{ V}}{5.8 \mu\text{s}} = 2.33 \text{ V } \mu\text{s}^{-1}$. This is much slower than the expected value.
- 16.79% Lost in Duty cycle.

Thank you!